

Test 28: f_GUE figure of merit

Verdict: FAIL | Generated: 2026-09-02 17:39:02 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

Statistical: fraction of spectral diagnostics consistent with GUE.

METHOD: Statistical: fraction of spectral diagnostics consistent with GUE.

Test 28 HARDCORE pass 2: 96x96, Nv=4, q=-1.0, W=1.0, n=5 realizations, per-realization f_GUE ensemble.

Test 28 HARDCORE: $\mu_{\text{GUE}} = 0.8553 - 0.0214$ (n=5), CI [0.8184, 0.8946], scatter $\sigma = 0.0479$.

Test 28b [HARDCORE pass 2]: 2 sub-checks, 1 failed ? WARN

Result

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/plot_02' (.png 600dpi / .svg / .pdf)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v19.jl --test 28 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Every computation is logged in logs/cor

Independent verification guide

[17:13:08.712] [+12303.994s] calc: ??(L=2.000) = 0.404688 (n_pos=9601, n_windows=3000, mean_count=1.99)

```
[17:13:09.363] [+12304.645s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 4 vortices, model=:monumental
```

FULL COMPUTATION LOG

[17.18:15.462] [+12610.7443] calc: ??(L=2.008) = 0.663437 (n_pos=5530, n_windows=3000, mean_count=2.03)

```
[17:18:19.349] [+12614.631s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 4 vortices, model=:monumental
```

[17:23:20.729] [+12916.011s] calc: ??(L=2.000) = 0.629287 (n_pos=5530, n_windows=3000, mean_count=2.02)

[17:23:26.725] [+12922.007s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 4 vortices, model=:monumental

[17:28:20.247] [+13215.529s] calc: ??(L=2.000) = 0.620146 (n_pos=5530, n_windows=3000, mean_count=2.04)

```
[17:28:26.188] [+13221.470s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 4 vortices, model=:monumental
```

[17:33:37.233] [+13532.515s] calc: ??(L=2.000) = 0.737871 (n_pos=5527, n_windows=3000, mean_count=2.04)

```
[17:33:43.480] [+13538.762s] build ab cloud hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 4 vortices, model=:monumental
```

[17:38:57.213] [+13852.495s] calc: ??(L=2.000) = 0.526840 (n_pos=5530, n_windows=3000, mean_count=2.02)

??

TEST 28 (pass 2, HARDCORE): f_GUE figure of merit ? ensemble deep pass

n ? 5 realizations, per-realization f_GUE, ensemble verdict

CONSOLE CAPTURE

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? RUNTIME: n=5 realizations at 96x96 ? expect ? 12?17 min

Config: 96x96, Nv=4, q=-1.00, ?=0.5000, W=1.0 (hardcore standard: W=1, q=1, Nv=4)

real 1/5: f_GUE = 0.8378

real 2/5: f_GUE = 0.8592

real 3/5: f_GUE = 0.8649

real 4/5: f_GUE = 0.7911

real 5/5: f_GUE = 0.9234

Ensemble: ?f_GUE? = 0.8553 – 0.0214 | bootstrap CI [0.8184, 0.8946] ? 0.90 (diagnostic)

??

?f_GUE? > 0.90 WARN ? ?f_GUE? = 0.8553 – 0.0214 (n=5 of 5 finite)

scatter < 0.10 PASS ? ?=0.0479 stable across vortex layouts

Test 28b [HARDCORE pass 2]: 2 sub-checks, 1 failed ? WARN